

**ADMINISTRATIVE INFORMATION****FMI No: 25403****FMI Title: HP80 Tube Pump Support Bracket**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☒	Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☐	Parts Ordering: GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐	No Parts are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems On-line Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

This FMI only applies to LightSpeed 5x HP80 systems with Varian tube and tube pump. If this system is an HP80 that was upgraded to an HP100, do NOT perform this FMI. Close your dispatch in code 2 - FMI not required.

Unit Tracked: 2357739-3, 2400465 and 2378661

FMI Instructions Part No.: 5241212-1EN

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
On-line Center
800-321-7937
Milwaukee, WI 53201, USA

GEMS - Europe

GEMS - E / ESC
BP34
F 78533 BUC CEDEX, FRANCE
Tel: (33 1) 39 20 00 07 or ESC
Fax: (33 1) 30 70 99 70

Asia Support Center
Phone: 81-426-56-0033
Fax: 81-426-48-2905

FMI No.: 25403

FMI Title: HP80 Tube Pump Support Bracket

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

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Effectivity

This FMI applies to the entire installed base of LightSpeed 5x HP80 systems with Varian tube and tube pump. This FMI is tracked by the HP80 Gantry Model Number (2357739-3, 2400465 and 2378661).

If the system on site is not an HP80 or is an HP80 that was upgraded to an HP100, do NOT perform this FMI. Close your dispatch in code 2 – FMI Not Required.

Purpose

The purpose of this FMI is to attach a support bracket to the inlet pipe of the oil pump. This bracket secures the inlet pipe and prevents the weld from cracking where the inlet pipe meets the pump body.

Prerequisite FMIs

None

Related FMIs

None

Furnished Materials

Qty	Description	Part #
1	FMI Document	5241212-1EN
1	Pump support bracket	5149939
1	Round bend u-bolt asm	5161681
1	Loctite 242	46-170686P1

Tools Required

Qty	Description	Part #
1	FMI Document	5241212-1EN
1	Field torque wrench kit	Field Supplied
1	5mm Allen wrench	Field Supplied
1	10mm and 11mm sockets with extension and 3/8" drive	Field Supplied

Personnel Requirements

"2" hours labor and "1" hour of travel

Before You Begin

Before beginning this procedure, please read the gantry safety information.

Safety



⚠ WARNING

POTENTIAL FOR SHOCK.

VOLTAGE MAY BE PRESENT.

USE PROPER LOCKOUT/TAGOUT PROCEDURES BEFORE REPLACING COMPONENTS.

Procedure

Preparation

1. Remove gantry right side cover.



NOTICE

Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

2. Disable axial drive and HVDC from the Service Switch Panel.
3. Rotate the gantry so the pump location is on the left (TGPU side) as shown in Figure 1.

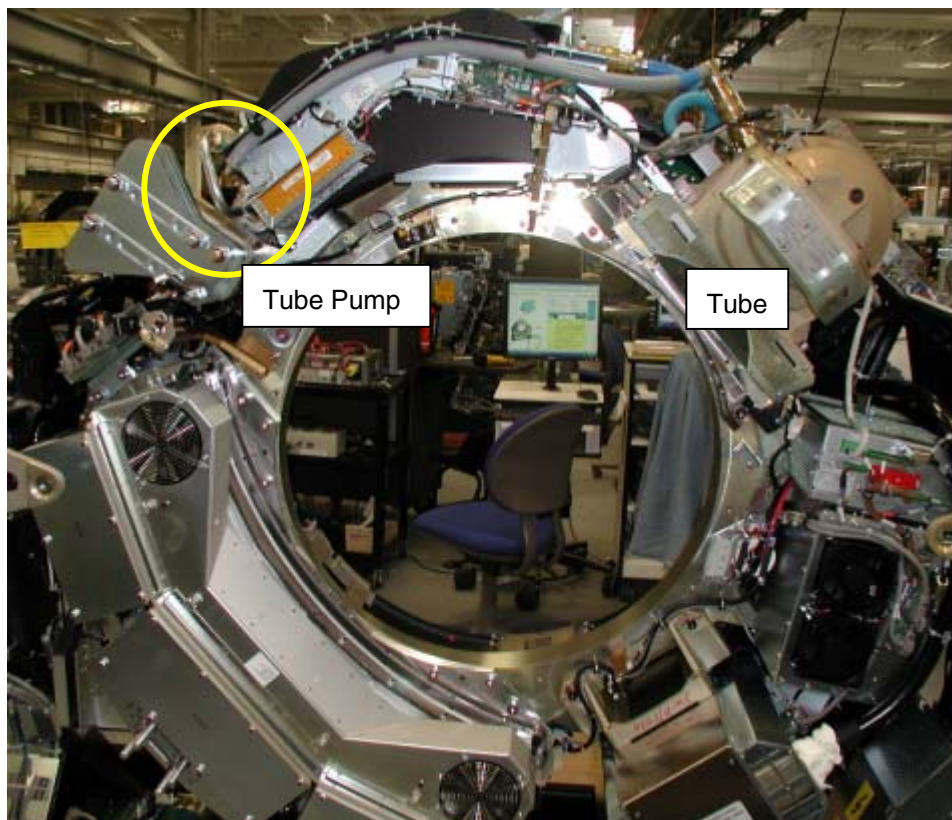


Figure 1. Gantry position

4. Shutdown system, Lock and Tag main power.
5. Remove left side, top and front gantry covers.

Bracket Installation Procedure

1. Use a 10mm socket and extension to remove the M6 nuts and washers from the two socket cap screws at the heat exchanger location shown in Figure 2. Leave the screws in place as they will be reused to mount the bracket. It may be necessary to use a 5mm Allen wrench on the backside to keep the cap screws from spinning.

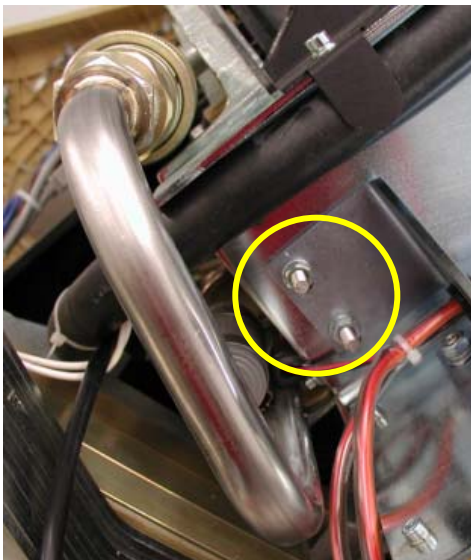


Figure 2. Heat exchanger screw location

2. Place the bracket (5149939) over the two screws on the heat exchanger and loosely install washers and nuts (Figure 3).



Figure 3. Installation of bracket over heat exchanger screws

3. Slide the bracket along the heat exchanger such that it makes contact with the oil pump inlet pipe (Figure 4).

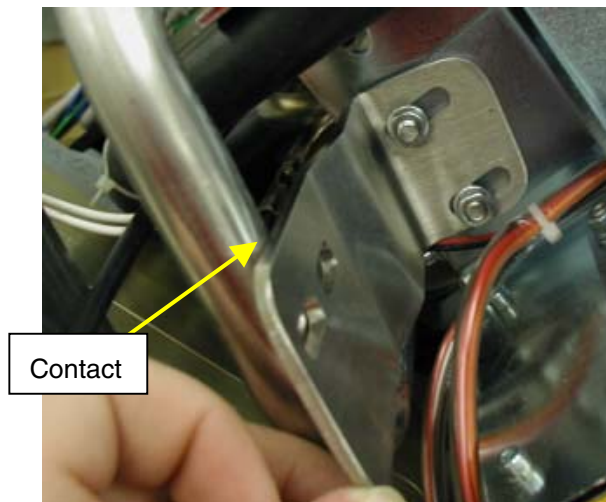


Figure 4. Positioning bracket in proper location

4. Hand tighten the two heat exchanger nuts, making sure no wires are pinched between the bracket and heat exchanger or tubing.
5. Disassemble the different parts of the u-bolt assembly (5161681). As shown in Figure 5, place the u-bolt around the pump inlet pipe and through the slots in the bracket. Install the washer plate over the ends of the u-bolt.

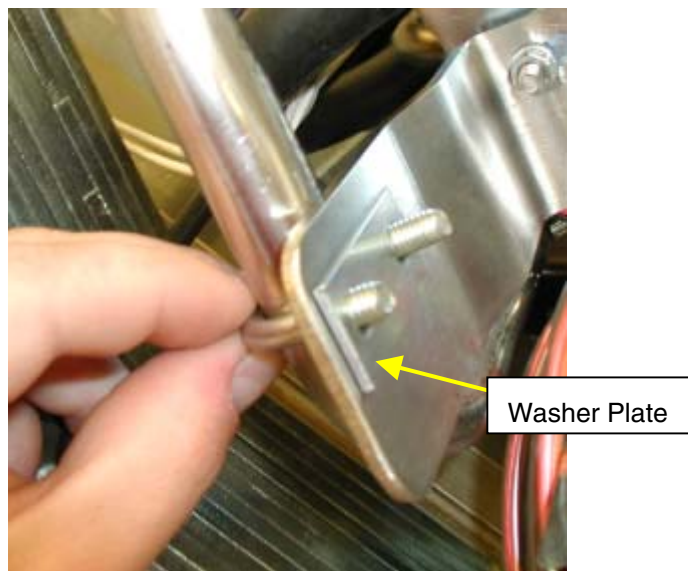


Figure 5. Installation of u-bolt and washer plate

6. Install and hand tighten the nuts on the ends of the u-bolt.

7. Inspect, again, that no wires are caught between the bracket and heat exchanger and tighten the two heat exchanger nuts using a 10mm socket with extension and a 5mm Allen wrench (Figure 6). Torque to the value in Table 1.

Table 1. Heat exchanger bolt torque

9.9 N-m	87.6 lb-in	7.3 lb-ft	101 Kg-cm
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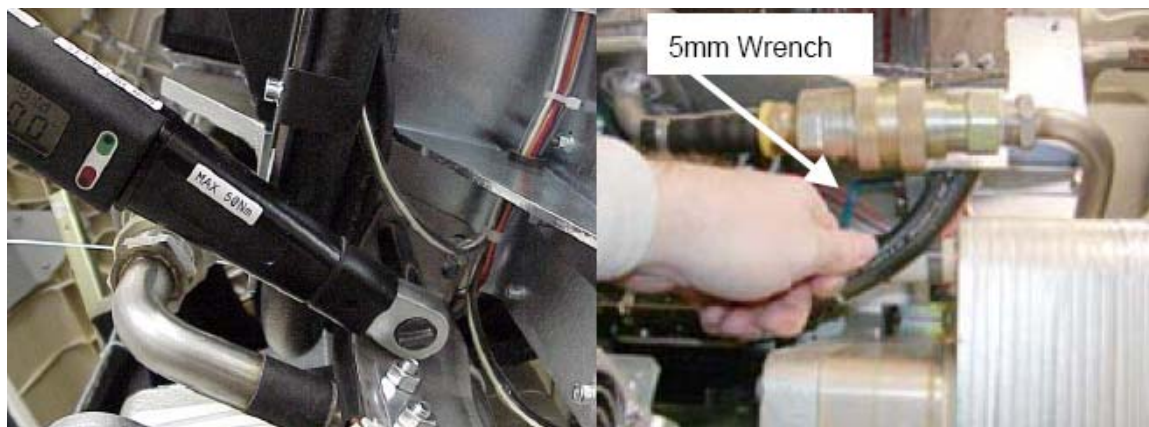


Figure 6. Torque heat exchanger mounting screws

8. Remove u-bolt nuts one at a time and apply Loctite 242. Tighten the nuts evenly such that there is about the same length of threads sticking out of the nuts. This ensures the u-bolt is not twisted or tilted and there is even pressure on the inlet pipe.
9. Torque the two u-bolt nuts to the value in Table 2. The completed installation is shown in Figure 7.

Table 2. U-bolt nut torque

8.0 N-m	70.8 lb-in	5.9 lb-ft	81.5 Kg-cm
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Figure 7. Completed installation of pump support bracket

Finalization

1. Rotate the gantry by hand and visually inspect that all components are securely installed.
2. Reinstall front, left side, and top gantry covers. Connect top cover fans.
3. Remove Lock and Tag. Turn on power to system.
4. Enable gantry power, HVDC, and Axial Drive from the Service Switch Panel. Enable table drive.
5. Reinstall right side gantry cover.
6. From the **Common Service Desktop, Calibration** menu, select [**Gantry Service Balance**] and perform a balance check. Rebalance gantry if necessary. The gantry should only require balancing if it was near the spec limits prior to adding the bracket.
7. Perform a test scan using the 20cm QA phantom to verify system operation.
8. Close the FMI per normal regional procedure and remove instructions from site.